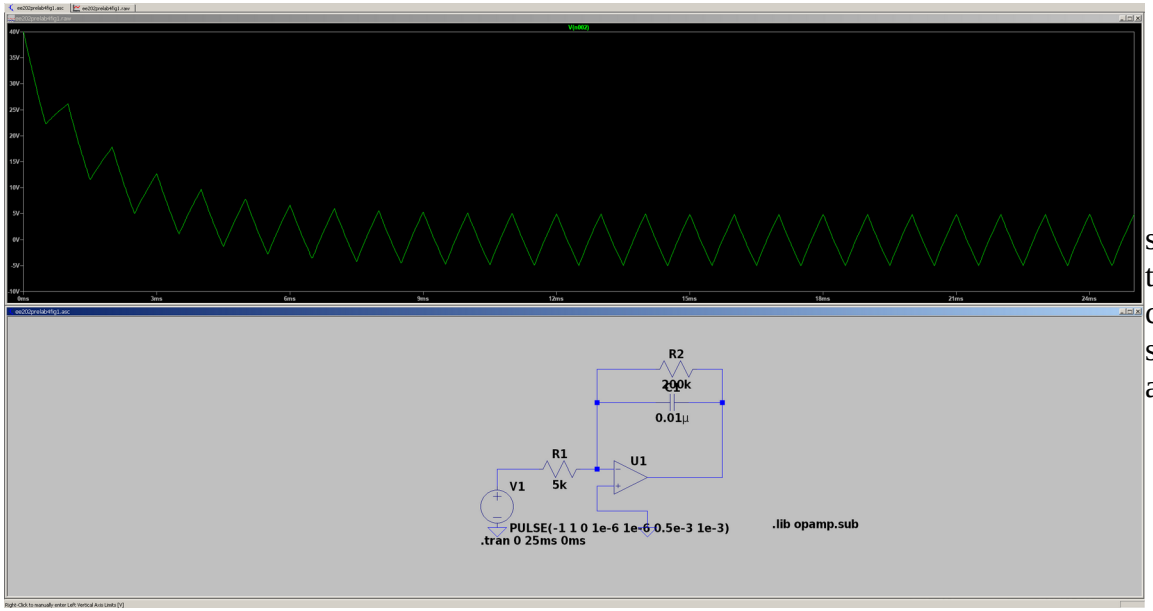
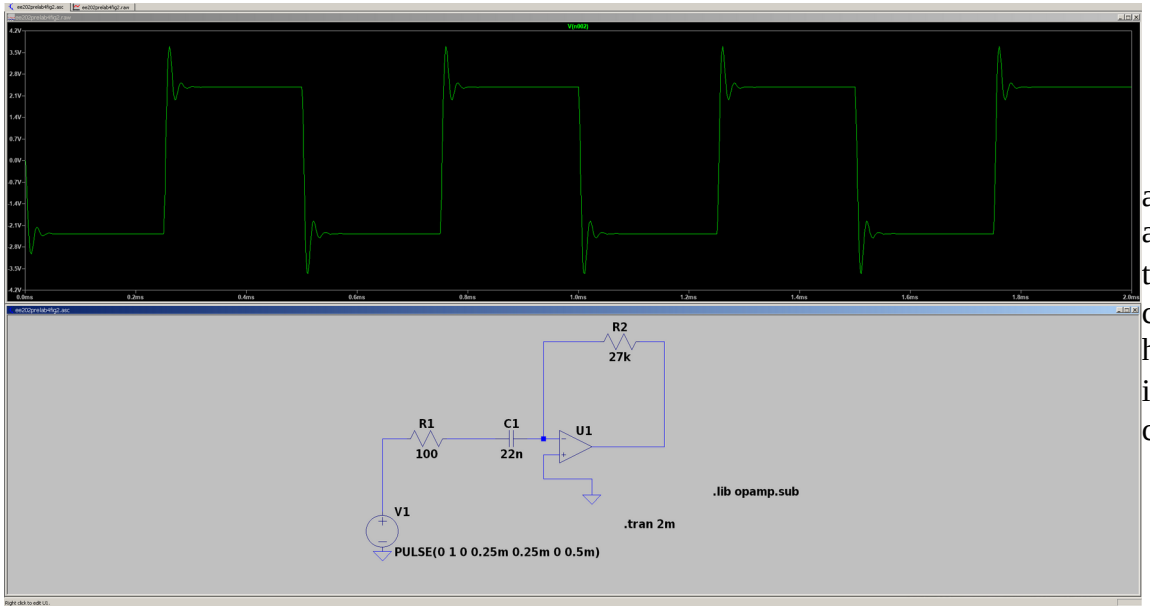


The simulation shows an initial 40V spike due to the circuit's starting DC conditions. However, due to the feedback resistor (R2), this initial offset diminishes away over the first 10 ms. Once the circuit settles, it successfully integrates the input, transforming the square wave into a stable, continuous triangular wave centered around 0V.



The simulation successfully demonstrates the differentiation of a 2kHz triangular wave into a square wave. The output voltage alternates between approximately +2.4V and -2.4V, which aligns with the theoretical mathematical derivative based on the R2 and C_1 time constant. Minor ringing is visible at the voltage transitions due to the high-frequency components at the sharp corners of the triangular input, though the series R1 resistor successfully keeps the overall circuit stable.